

Immigration, Gender, and Citizenship: Intersecting Inequalities in the U.S. Labor Market

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Abstract:

This study examines wage disparities based on gender and citizenship status. A Mincer earnings regression and Blinder-Oaxaca decomposition were utilized to estimate wage differentials while controlling for human capital characteristics, sociodemographic factors, and industry and geographic fixed effects. Results indicate wage gaps persist even after robustness checks: females earn approximately 32.7% less than males, primarily due to intangible factors such as employer bias and statistical discrimination. The analysis further reveals that non-citizen women face an additional 7.9% wage penalty beyond the independent effects of gender and citizenship alone. These findings reveal unfair treatment of immigrants, women, particularly non-citizen women, calling for intersectional policy reform.

Keywords: Wage gap · Citizenship · Gender wage differential · Immigrant · Intersectionality

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I. Introduction

It is often said that when the U.S. catches a cold, Mexico gets pneumonia – a reflection, valid or not, of the economic interdependence between Mexico and the U.S. (Malkin, 2008). Yet, is this true? Is immigration truly detrimental to the U.S. economy? If so, why has the U.S. implemented policies such as the Bracero Program, which allowed Mexicans to work in the U.S. for up to six months? (US House of Representatives: History, Art & Archives, n.d.; Massey et al., 2002). The truth is that there is nothing more permanent than a temporary work permit.

Today, the foreign-born population in the United States stands at 47.8 million, representing 14.3% of the total population (Pew Research, 2024). Of these immigrants, Mexican immigrants comprise the largest group at 23% of all U.S. immigrants (Pew Research, 2024). Among the total immigrant population, approximately 11 million are unauthorized (Baker and Warren, 2022). Despite the immigrants' role in the U.S. economy, particularly in labor-intensive industries, they often face significant barriers and are perceived by policymakers as what I will hereafter term an “undesired necessity” (National Immigration Forum, 2017).

The debate over immigration's economic impact remains contentious among economists. The Mariel boatlift of 1980, which brought 120,000 low-skilled Cuban migrants to Miami, exemplifies this disagreement. While the National Academies of Sciences (2017) found little evidence that this influx harmed native workers' wages or employment, Borjas (2017) argued it significantly depressed low-skilled native wages. Card and Peri (2016) later challenged these findings, concluding there was “essentially no effect” on Miami's labor market.

Rather than engage this broader debate, this paper focuses on a critical yet understudied aspect: the intersection of citizenship status and gender in wage determination. Specifically, it will examine how these factors shape wage disparities among foreign-born workers in the agriculture, construction, and service industries—industries heavily reliant on immigrant labor,

both authorized and unauthorized. With this research, I hope to contribute to the following quagmire: *How do citizenship status and sex shape wage disparities among foreign-born workers in the agriculture, construction, and service industries?* This paper analyzes individual-level data from the 2012–2019 American Community Survey (ACS). Geographic diversity is ensured through a 5% stratified random sample of Public Use Microdata Areas (PUMAs). I estimate wage determinants with a Mincer regression and quantify discrimination through a Blinder–Oaxaca decomposition. The remainder of the paper proceeds as follows: Section II reviews the literature, Section III presents the theory, Section IV describes the data, Section V outlines the empirical strategy, Section VI reports results, and Section VII concludes.

II. Literature Review

While there is a dispute on immigration’s broader impact on the U.S. economy, there is consensus on one observation: a wage gap between immigrant and native workers (Borjas, 1994; Coulombe et al., 2014; Smith & Fernandez, 2017). Scholars constantly debate the relative weight of the following three wage determinants:

1. Human Capital

Immigrants’ wage disadvantages are often attributed to disparities in human capital. Borjas (1994) argues that U.S.-born workers inherently benefit from English proficiency, which allows immigrants to escape “ethnic enclaves” and access broader opportunities (Garg, 2004). Education also plays a critical role: immigrants complete 1.3 fewer years of formal education than natives and are far more likely to lack a high school diploma (Camarota & Zeigler, 2016; Card, 2005). Even when immigrants possess comparable qualifications, foreign credentials are systematically devalued. Chiswick (1978) found that immigrants receive lower returns on foreign education and experience, a pattern Blau and Kahn (2015) confirmed, showing U.S. employers discount “non-domestic skill signals” for newly legalized immigrants.

However, human capital alone cannot fully explain persistent gaps. Coulombe, Grenier, and Nadeau (2014) find that once human capital quality is accounted for, the unexplained portion of the immigrant–native wage gap decreases (by 62% for men and is nearly eliminated for women). Donahue (2021) reinforces this critique, finding that wage gaps persist even among immigrants with 25+ years of U.S. residency, suggesting structural barriers like unfair labor policies or discrimination. Coulombe et al.’s study is used as a conceptual guide rather than as an empirical precedent for the U.S., as their study is based in Canada.

2. Occupational Segregation

Immigrants face systemic barriers that funnel them into low-wage industries. Borjas (2018) attributes this to “imperfect substitutability,” arguing that immigrants cluster in occupations natives avoid. Equally, Smith and Fernandez (2017) found that immigrants disproportionately work in low-wage sectors, while Garg (2004) emphasizes how limited English proficiency traps them in industries like construction, where wages lag by 12–18%.

Occupational segregation is not voluntary. Elliott and Lindley (2008), Blau and Kahn (2015), and Ottaviano and Peri (2012) argue that limited education restricts immigrants to unskilled jobs, but this ignores gender dynamics. Vernez (1999) notes that female immigrants face “additional cultural and social barriers,” such as childcare responsibilities and spousal income effects (Schoeni, 1998). A husband’s higher wages may reduce a woman’s labor force participation (income effect) or increase her household productivity (substitution effect).

3. Discrimination

Discrimination is a persistent yet untackled factor. Oreopoulos (2011) attributes wage disparities to “various intangible factors,” including employer bias. Hersch (2011) quantifies colorism, showing that immigrants with darker skin tones earn 16-23% less than lighter-skinned

peers, indicating that wage gaps persist beyond human capital and reflect racialized judgments similar to those tied to non-citizen status. These penalties intersect with gender. Vernez (1999) documents compounded barriers for immigrant women, and Garg (2004) shows Latina workers concentrated in the sub-\$35,000 bracket compared to Asian counterparts. This highlights the persistent intersection of racist and sexist biases in economic outcomes and research.

Citizenship status increases these inequities further. The National Academies of Sciences, Engineering, and Medicine (2017) stress that unauthorized workers face heightened vulnerability, yet no study compares naturalized and non-citizen women's exposure to discrimination. Massey et al. (2002) and the National Immigration Forum (2017) emphasize industries like agriculture, where non-citizens dominate, but they fail to analyze how citizenship-gender intersections shape outcomes.

Henceforth, existing research leaves critical gaps that this paper aims to address directly. First, while scholars like Borjas (1994) and Blau and Kahn (2015) emphasize credential devaluation, no work examines whether naturalization improves returns to foreign education in key industries like service, agriculture, or construction. Second, though Vernez (1999) and Garg (2004) identify gendered barriers for female immigrants, they overlook how citizenship status—particularly the distinction between naturalized and non-citizen women—shapes occupational mobility out of “ethnic enclaves” (Garg, 2004). Third, discrimination frameworks (Hersch, 2011; Oreopoulos, 2011) treat immigrants as uniform groups, failing to capture how non-citizen women face compounded penalties distinct from naturalized peers.

This study tackles these gaps by disaggregating citizenship status to test whether legal protections moderate human capital. It also models intersectional penalties through gender-citizenship-industry interactions, building on Donahue's (2021) finding that “structural

problems” perpetuate disparities even among equally qualified workers. A Blinder-Oaxaca decomposition isolates discrimination’s role, extending Oreopoulos’ (2011) focus on “unobserved factors” to quantify bias in non-citizen women’s wage penalty. Hence, this paper hypothesizes that, *ceteris paribus*, female immigrants will earn less than (a) male immigrants, (b) native males, and (c) native females within these industries. By focusing on high-immigrant industries, this paper shows how sectoral dynamics shape inequality, addressing the National Academies’ (2017) call for intersectional analysis.

III. Economic Theory

Human capital theory (Becker, 1964; Mincer, 1974) posits that wages reflect investments in education, skills, and experience. However, immigrants face credential devaluation, where foreign education and work experience are systematically discounted by employers (Chiswick, 1978; Blau and Kahn, 2015).

Taste-based discrimination (Becker, 1957) extends this by framing employer prejudice as a willingness to sacrifice profits to avoid hiring certain groups. Employers act as if hiring non-citizens, women, or both imposes a nonpecuniary cost, quantified by a discrimination coefficient d . The perceived cost of employing a member of a disfavored group thus becomes $w(1 + d)$, exceeding their market wage unless the worker accepts a discount.² Credential devaluation exacerbates this: foreign degrees, especially from non-Western institutions, are treated as noisy or unreliable signals of productivity (Banerjee and Lee, 2015; Li, 2001). Donahue (2021), for example, finds that immigrants with foreign credentials earn 18% less than natives with comparable U.S. qualifications, even after controlling for occupation.

² In Becker’s model, w is the worker’s market wage and d is the discrimination coefficient. If $d > 0$, the employer behaves as though the cost of hiring the worker is $w(1 + d)$. For example, if $w = \$20/\text{hour}$ and $d = 0.2$, the employer perceives the cost as $\$24/\text{hour}$, making the worker less competitive unless they accept a wage discount.

Based on both Human Capital Theories and Taste-Based Discrimination, I hypothesize that non-citizen women will exhibit systematically lower returns to education than their naturalized or native-born peers, particularly in agriculture, where credential devaluation is strong and recognition is limited.

Statistical discrimination complicates this dynamic. Employers use group averages, based on citizenship status and/or gender, to proxy unobserved productivity under imperfect information (Aigner and Cain, 1977). Non-citizens may be stereotyped as unreliable (visa instability) and women as less committed due to caregiving roles, compounding penalties for non-citizen women.³ Based on this new theoretical framework of Statistical Discrimination, I expect that unexplained wage gaps (via Blinder-Oaxaca) will be largest for non-citizen women, reflecting employer reliance on group stereotypes rather than individual productivity.

While taste-based discrimination reflects explicit prejudice, statistical discrimination rationalizes bias through perceived risk—yet both converge in devaluing human capital.

Immigration policies create institutional barriers for non-citizens, limiting labor protections and access to high-wage sectors (Massey et al., 2002). Unauthorized workers lack minimum wage protections in agriculture, while temporary visa holders remain dependent on employers. Naturalization mitigates these barriers by granting legal protections. However, non-citizens—especially women—remain concentrated in industries with weak enforcement of domestic work, seasonal demand (agriculture), and services. While unionized sectors may offer narrower wage gaps through collective bargaining, non-citizen women are often excluded from membership (National Immigration Forum, 2017). Thus, Institutional Theory suggests that

³ For example, if employers statistically associate unauthorized status with job turnover, they may avoid investing in training for non-citizen women, relegating them to informal roles (Massey et al., 2002).

naturalized citizens likely experience narrower wage gaps compared to natives, while non-citizens face persistent penalties, especially in informal sectors.

Intersectionality theory argues that overlapping identities create multiplicative, not additive, disadvantages (Browne and Misra, 2003). Non-citizen women face “double jeopardy”: credential devaluation (human capital) and stereotypes about caregiving roles (statistical discrimination). For instance, Latina immigrants are often channeled into low-wage care work, where employers assume they will tolerate subminimum wages (Garg, 2004).⁴ This intersectional framework suggests non-citizen women face wage penalties exceeding the combined effects of gender and citizenship status alone.

Hence, by considering these four frameworks, I set up wage as a function of:

$$Wage = f(\text{Human Capital, Discrimination, Institutional Barriers, Intersectional Identity}) \quad (1)$$

My hypotheses, thereby, are the following:

H_0 : *There is no wage gap between non-citizen women in secondary sectors and their counterparts, once human capital differences are accounted for.*

H_A : *There might be a wage gap between non-citizen women in secondary sectors and their counterparts, once human capital differences are accounted for.*

IV. Data Description

This study analyzes individual-level American Community Survey (ACS)⁵ data from the years 2012 up to 2019, a period selected to ensure consistency in Public Use Microdata Area (PUMA) boundaries (Ruggles et al., 2025). The raw dataset comprises 25.3 million observations. After applying a 5% stratified random sample of PUMA-state combinations, removing

⁴ This illustrates Becker’s foundations, $w(1+d)$. A higher d inflates the perceived cost of hiring, leading employers to offer lower wages or exclude Latinas entirely.

⁵ Obtained from the Integrated Public Use Microdata Series (IPUMS). See Ruggles et al. (2025) for data source.

duplicates, and cleaning invalid entries (e.g., errors in income values, and ages outside the working-age range of 16–66), the final sample includes 526,832 individuals and 118 unique geographic identifiers. Geographic consistency is maintained through state FIPS codes, and the panel structure employs census tract fixed effects to account for localized economic conditions.

The dataset captures variables for analyzing labor market inequalities: citizenship status (categorized as citizen, naturalized citizen, or non-citizen), sex, industry, and wage income (adjusted in 2015 dollars)⁶. This study defines *immigrants* as individuals born in a foreign country. Naturalized citizens are categorized separately to examine how the naturalization process may influence wage disparities. [Table 1](#) shows that 85.39% of the sample are U.S.-born citizens, while 14.61% are foreign-born (7.96% naturalized, 6.65% non-citizens). Gender distribution is nearly equal (50.58% male, 49.42% female), though wage disparities are notorious: men earn an average of \$62,537 compared to women's \$41,441. These averages reflect selection into the full-time, wage-earning labor force captured by the ACS sample.

[Table 2](#) gives us the first glimpse of the differences in annual mean wages of each group. Notably, naturalized citizens earn higher mean annual wages (\$58,556) than native-born citizens (\$52,517), suggesting positive selection into naturalization. Second, non-citizens—a group likely to include unauthorized workers—have the lowest mean wages (\$41,713) and exhibit a pronounced gender gap: non-citizen women earn just 66% of their male counterparts' wages (\$31,864 vs. \$48,422).

Non-citizen workers are notably concentrated in agriculture, construction, and key service industries. [Table 3](#) shows that service sectors alone absorb a substantial share: Professional Services accounts for 13.77%, Arts for 13.26%, and Other Services for 5.51% of

⁶ All wages and household income are first converted to 1999 dollars via IPUMS's CPI99, then adjusted to 2015 dollars by multiplying by 1/0.703, where 0.703 is the CPI99 index value for 2015.

non-citizen employment. Agriculture & Forestry employs 2.55%, while Construction represents 10.08%. Together, these five sectors comprise 45.17% of all non-citizen workers. Figure 1 further illustrates that both naturalized and non-citizen workers are disproportionately represented in labor-intensive, and lower-wage industries such as agriculture, construction, and professional services.⁷

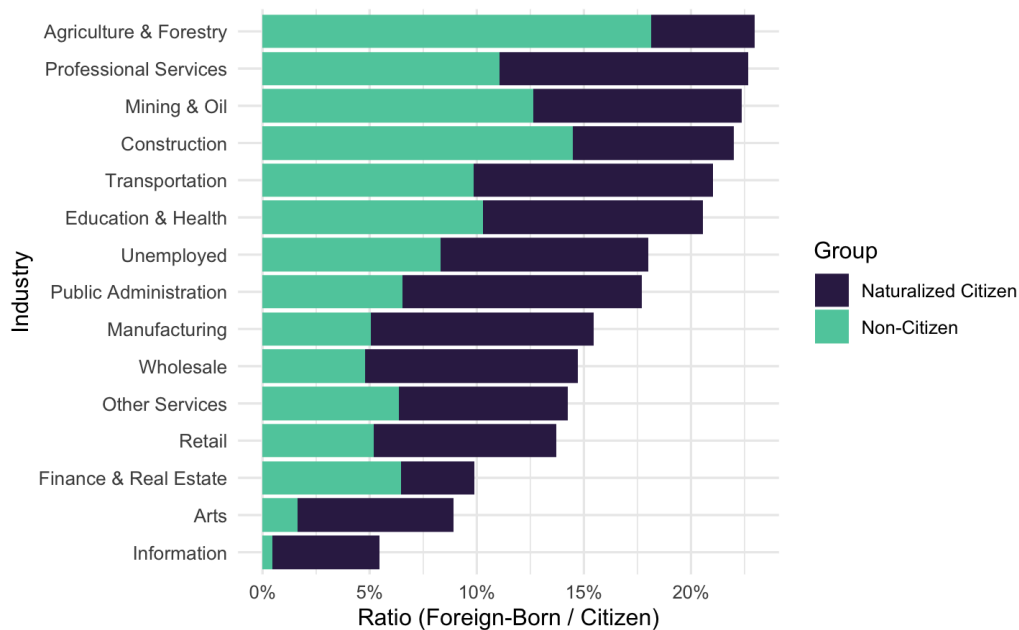


Fig. 1: Ratio of Foreign-Born to Citizen Workers by Industry

The dataset also reflects demographic diversity of the foreign-born population. As shown in Figure 2, most immigrants originate from Mexico, Southeast Asia, and Europe. Birthplace groupings were defined in accordance with the ACS codebook suggestions. For example, Mexico was assigned its own category—separate from other Latin American regions—due to its large representation in the dataset.

⁷ Ratios were calculated by dividing the number of foreign-born workers in each industry by the number of U.S.-born citizen workers in the same industry.

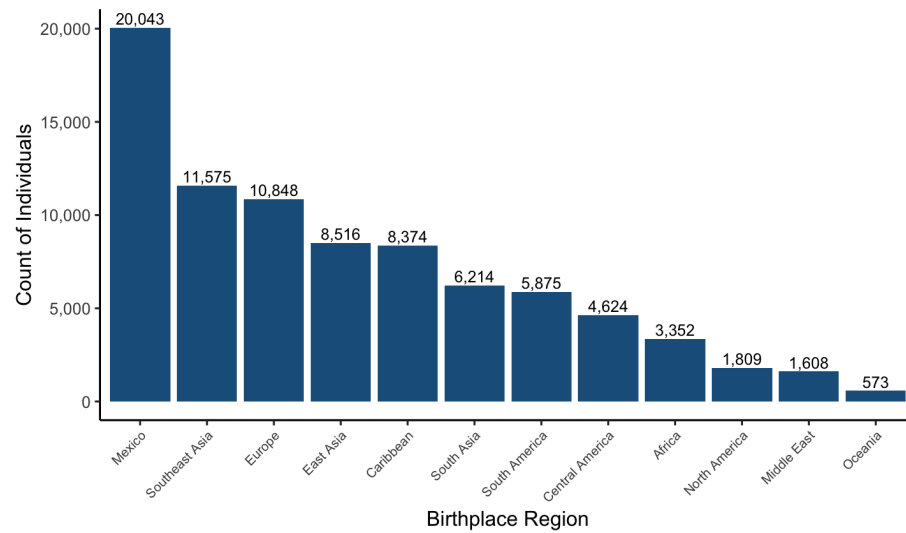


Fig. 2: Demographic Diversity by Birthplace

Figure 3 displays the distribution of income levels within educational groups, disaggregated by gender and citizenship. Even among the highly educated, non-citizen women remain disproportionately represented in the low-income bracket. By contrast, native-born men show the most favorable income distribution, especially at higher educational levels.

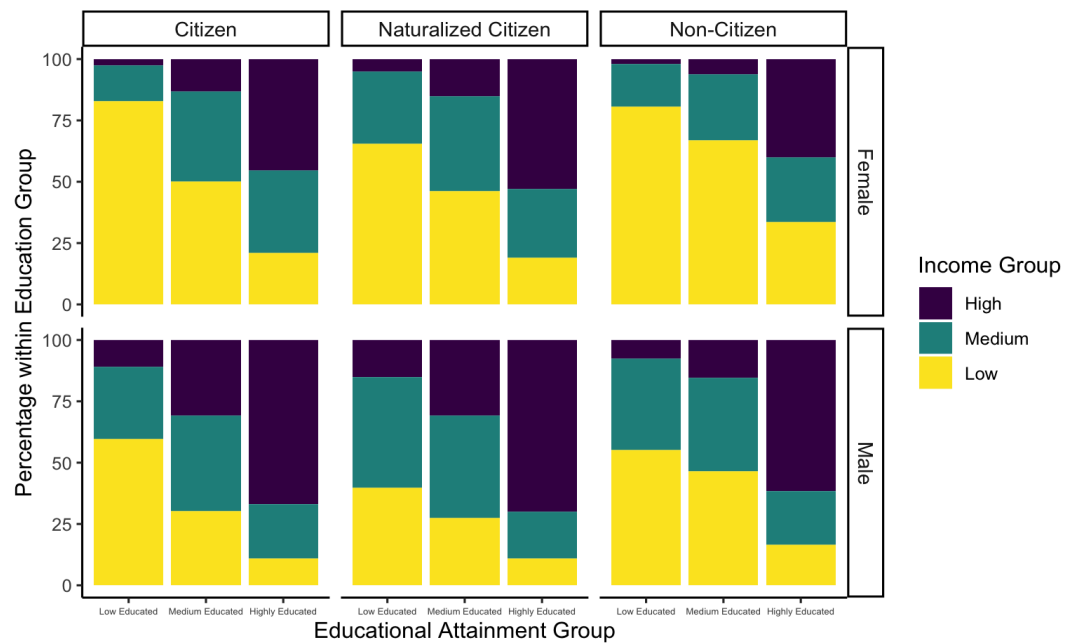


Fig. 3: Income Group Distribution by Gender, Citizenship, and Education

V. Empirical Strategy

I utilize two econometric approaches to explore the wage differentials between citizenship status and sex, focusing on teasing out the role of discrimination:

A. Mincer Earnings function

I begin with a baseline Mincer wage equation to capture the returns to human capital and other observable characteristics:

$$\ln(wage_{i,s,t}) = \beta_0 + \beta_1 Citizen_{i,s,t} + \beta_2 Sex_{i,s,t} + \beta_3 (Citizen * Sex)_{i,s,t} + X'_{i,s,t} \beta_4 + \quad (2)$$

$$\sum_{s=1}^S \gamma_s D_{s,i} + \sum_{t=1}^T \tau_t D_{t,i} + \sum_{j=1}^J \theta_j D_{j,i} + \sum_{k=1}^K \delta_k D_{k,i} + \epsilon_{i,s,t}$$

where, $\ln(wage_{i,s,t})$ is the log of the wage individual i ; in geographic area s (in our case, the PUMA–state combination); and t indexes the year when the wage is observed. Although industry effects are incorporated via fixed effects – j , the wage variable itself is recorded at the individual level. Here, each β is estimated. β_0 captures the baseline wage for the reference group, U.S. native males. The coefficients β_1 and β_2 capture the effects of citizenship status and sex, respectively, indicating how each factor influences wages. Lastly, β_3 is our interaction term, which allows us to assess whether the effect of citizenship on wages varies by sex.

The vector $X'_{i,s,t}$ bundles human capital variables—such as education, age, age squared⁸, household income, number of children, and English proficiency—with their associated coefficients β_4 , controlling for individual productivity-related factors that might otherwise confound our estimates. The summation terms incorporate fixed effects for PUMA–state (γ_s),

⁸Age and age squared were centered at its sample mean to reduce multicollinearity (all VIFs < 2.5), so that the age coefficients represent effects at the average age rather than at zero.

year (τ_t), industry (θ_j), and birthplace (δ_k), absorb unobserved geographic, temporal, and sectoral heterogeneity, isolating within-group variation in wages. Lastly, $\epsilon_{i,s,t}$ accounts for all unobserved factors that may influence wages but are not explicitly modeled.

B. Blinder-Oaxaca decomposition

I then apply a Blinder–Oaxaca decomposition to four dyadic contrasts: 1) the overall gender gap, 2) the citizenship gap, 3) the gap between non-citizen and citizen women, 4) and the gap between non-citizen women and non-citizen men. This method splits wage gaps into two components, explained and unexplained, allowing to isolate structural barriers—such as discrimination or unequal returns to human capital—that persist even after accounting for observable differences. The Blinder-Oaxaca decomposition equation is structured as follows:

$$\overline{\ln(wage)}_A - \overline{\ln(wage)}_B = (\bar{X}_A - \bar{X}_B)\hat{\beta}_B + \bar{X}_A(\hat{\beta}_A - \hat{\beta}_B) + (\bar{X}_A - \bar{X}_B)(\hat{\beta}_A - \hat{\beta}_B) \quad (3)$$

Here, the bars represent group sample mean. Group A represents the disadvantaged cohort (women, non-citizen, or both), and Group B the advantaged comparator. The explained component, $(\bar{X}_A - \bar{X}_B)\hat{\beta}_B$, reflects wage gaps attributable to differences in observable characteristics (such as education or English proficiency). The unexplained component, $\bar{X}_A(\hat{\beta}_A - \hat{\beta}_B)$, captures disparities in returns to these traits (lower pay for identical qualifications), interpreted as structural discrimination or unobserved barriers. Finally, the interaction term, $(\bar{X}_A - \bar{X}_B)(\hat{\beta}_A - \hat{\beta}_B)$, accounts for simultaneous differences in endowments and returns.

I acknowledge that the non-citizen category includes both unauthorized workers and temporary visa holders like international students and H-1B/H-2A workers,⁹ who differ in legal rights and labor market access. Despite my model including numerous human capital control variables and fixed effects, unobserved factors—such as ability, visa type nuances, social networks, health status, or union membership—may still bias the estimates. Equally, the Oaxaca “unexplained” component, while serving as a proxy for discrimination, also captures residual heterogeneity. Thus, my results should be interpreted not as definitive causal effects, but as approximations.

Building on Oreopoulos (2011) and Donahue (2021), this approach uses a Mincer regression to model baseline wage determinants and a Blinder-Oaxaca decomposition to quantify how much of the remaining gap, particularly for non-citizen women, reflects structural barriers like discrimination.

VI. Results

A. Regression Results

Table 4 presents the regression results examining the effect of gender and citizenship on log wages across three model specifications. Each model progressively provides refined estimates by controlling for individual characteristics and structural factors.

The first regression captures the raw associations between citizenship, gender, and wages without accounting for human capital or structural factors. The results show that naturalized citizens earn 14.8% more than natives, while non-citizens earn 24.7% less, reflecting unadjusted

⁹ H-1B visas are granted to high-skilled specialty workers, typically to international students after receiving their college degree. Generally, H1-B lasts a year, but for STEM fields it can last up to 3 years. H-2A visas are issued to temporary agricultural workers for seasonal farming jobs. Since the abolition of the Bracero Program, temporary agricultural workers have been admitted with H-2A visas.

structural and human capital disparities. Women face a 40.3% wage gap, underscoring systemic gender inequality. Non-citizen women suffer an additional 4.0% penalty beyond the separate effects of gender and citizenship. Although this model is very simple, it still provides initial evidence of intersectional disadvantage for non-citizen women.

Table 4: Effect of Gender and Citizenship on Log Wages

VARIABLES	Model 1	Model 2	Model 3
Naturalized Citizen (NC)	0.148*** (0.007)	-0.007 (0.006)	0.037** (0.017)
Not a citizen (NotC)	-0.247*** (0.007)	-0.010 (0.006)	0.004 (0.018)
Female	-0.403*** (0.003)	-0.398*** (0.003)	-0.327*** (0.010)
Naturalized Citizen* Female	0.019* (0.010)	0.022*** (0.008)	0.010 (0.018)
Not Citizen * Female	-0.040*** (0.012)	-0.090*** (0.010)	-0.079*** (0.015)
Constant	10.583*** (0.002)	9.761*** (0.008)	9.767*** (0.015)
Observations	526,832	526,832	526,832
Controls		X	X
Fixed Effects			X
R-squared	0.038	0.425	0.460

Robust Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Model 1: Captures raw associations between citizenship, gender, and wages. Does not account for human capital.

Model 2: Controls for education, age, and other individual human capital characteristics.

Model 3: Adds fixed effects for industry, year, birthplace, and geography. Robust standard errors are clustered by 118 combined state-PUMA units (FIPS × PUMA).

Model 2 shows that the disparities largely vanish, confirming that the gaps in Model 1 stem from differences in education, skills, and experience. Importantly, the disappearance of the non-citizen penalty occurs in Model 2, before fixed effects, showing that human capital controls drive this change. The gender penalty persists at 39.8%, highlighting discrimination's entrenched role. There is an interaction shift where the penalty reduces for naturalized women (+2.2%

premium), suggesting human capital offsets bias for this group. The penalty worsens for non-citizen women (-9.0%), revealing that structural barriers (e.g., credential devaluation, occupational segregation) exacerbate their disadvantage. Lastly, the third regression adds fixed effects for industry, state, birthplace, and year. This specification controls for unobserved heterogeneity across geographic locations, industries, and time periods that might otherwise confound our estimates. More importantly, it isolates the pure effects of citizenship and gender within specific contexts and strengthens inference through cluster-robust standard errors.

Results reveal a 3.7% wage premium for naturalized citizens ($p < 0.05$), aligning with Borjas (1994) and Donahue (2021), who argue that legal status mitigates structural barriers but does not fully erase disparities. The absence of a significant non-citizen penalty contrasts with Coulombe et al. (2014) and Smith & Fernandez (2017), who emphasize human capital deficits, suggesting that within specific industries/locations, non-citizens' outcomes may reflect sectoral dynamics (e.g., unionization in construction vs. informality in agriculture) as noted by Massey et al. (2002). The persistent 32.7% gender penalty ($p < 0.01$) across specifications underscores systemic devaluation of women's labor, consistent with Hersch (2011) on colorism and Garg (2004) on gendered occupational segregation. Critically, the "double disadvantage" for non-citizen women (-0.079 , $p < 0.01$) reflects intersectional discrimination theorized by Browne & Misra (2003), where citizenship and gender multiplicatively disadvantage workers. This aligns with Vernez (1999), who identifies cultural and institutional barriers for immigrant women, and Oreopoulos (2011), who attributes unexplained wage gaps to employer bias.

Selection effects are likely driving much of our estimated gaps. For instance, naturalized citizens must pass English and civics tests and maintain steady employment, so those who naturalize are a positively selected, higher-ability group. This selection inflates the raw

citizenship premium in Model 1 (14.8%) but largely vanishes once we control for education, age, and fixed effects (3.7%). Conversely, the “non-citizen” category includes both unauthorized workers in low-wage, high-risk jobs and H-1B or H-2A visa holders in higher-paid sectors, so estimated penalties blend actual returns with who sorts into those visa classes.

These findings align with the theoretical framework of statistical discrimination outlined earlier. Employers may use citizenship status as a proxy for unobserved productivity, particularly for women. The persistence of the negative interaction term for non-citizen women reveals how employer bias amplifies their wage disadvantage.

B. Blinder-Oaxaca Decomposition

This method allows me to decompose wage gaps into three components: (1) explained, reflecting differences in observable characteristics; (2) unexplained, capturing differences in returns to those characteristics; and (3) interaction effects between explained and unexplained. The unexplained component is particularly important as it often serves as a proxy for discrimination or unobserved structural barriers.

The Blinder-Oaxaca decomposition¹⁰ results reveal a multi-dimensional narrative of wage disparities shaped by gender, citizenship, and their intersection. Gender-based gaps are overwhelmingly driven by systemic discrimination, with 98.7% of the 39.7% wage difference between men and women attributable to unequal returns to identical qualifications—mostly seen in education, where women receive -0.254 lower returns than men. In contrast, citizenship gaps (24.3%) stem primarily from structural inequalities: 85.6% of the disparity reflects non-citizens’ lower educational attainment and economic resources, rather than unequal pay for equivalent traits. However, intersectionality magnifies disadvantages: non-citizen women face a “double penalty”, earning 30.2% less than citizen women due to both human capital deficits and

¹⁰ For more information on each decomposition check the Appendix: [Detailed Blinder-Oaxaca Analysis](#)

discriminatory returns to education (-0.454). Most critically, non-citizen women experience the largest disparity (-44.3%) compared to non-citizen men, despite equivalent or better qualifications, as 107.4% of this gap arises from systemic undervaluation of their traits. These findings underscore how citizenship status and gender interact—non-citizens contend with structural barriers, while women, especially non-citizens, face compounded discrimination, aligning with theories of statistical discrimination where employers devalue femininity and foreign status as proxies for productivity.

To better understand how wage disparities vary by method, [Figure 4](#) compares results from the Mincer regressions and the Oaxaca decomposition. While Mincer captures total wage differentials, Oaxaca helps distinguish between disparities due to observable characteristics and those due to differential returns—often interpreted as structural discrimination.

Fig. 4: Mincer Regression and Oaxaca Decomposition

Category	Mincer Regression	Oaxaca Decomposition
Purpose	Estimates overall association between predictors and log wages	Decomposes wage gap into “explained” vs. “unexplained” parts
Focus	Magnitude of wage differentials by group (e.g., women vs. man, citizens vs. non-citizens)	How much of that differential is due to characteristics vs. differential returns
Gender Result	Women earn ~33% less than men, even after controls	~90% of this gap is unexplained, suggesting structural inequality
Citizenship Result	No significant wage gap remains after controls; raw gaps reflect selection and human capital	Among non-citizens, returns to education are far lower
Limitations	Does not isolate source of gap; all effects bundled	Unexplained part may include omitted variables, not just discrimination

Returning to the hypotheses introduced in the theoretical section, the evidence rejects the null hypothesis that non-citizen women experience no wage gap once human capital differences are accounted for. Across all specifications, non-citizen women consistently exhibit statistically significant wage penalties, even after controlling for human capital, sectoral composition, geography, and time. Thus, the results support the alternative hypothesis: non-citizen women face meaningful wage disparities that cannot be explained solely by observable characteristics.

VI. Conclusion and Discussion

Immigrants are economically indispensable yet constantly undercompensated by the U.S. labor market. The results highlight a troubling reality: citizenship status and gender intersect to exacerbate wage disparities, impacting non-citizen women the most. While human capital factors largely explain wage differences for non-citizens, a significant 32.7% gender gap persists even after controls—highlighting the structural devaluation of women’s labor. Echoing Oreopoulos (2011), these disparities appear driven largely by intangible factors such as employer bias and statistical discrimination, where gender acts as a proxy for unobserved productivity (Aigner & Cain, 1977).

The initial wage penalty for non-citizens vanishes after introducing human capital controls, aligning closely with Coulombe et al. (2014), who attribute gaps primarily to differences in education, skills, and experience rather than overt discrimination. However, the modest but significant naturalization premium of 3.7% reinforces Donahue (2021) and Borjas (1994), indicating that legal status improves labor market outcomes by mitigating, though not eliminating, structural barriers. Furthermore, the premium suggests tangible economic benefits from naturalization, such as enhanced labor market access and greater legal protections, as

highlighted by Blau and Kahn (2015). Nevertheless, naturalization alone does not fully close the persistent disparities rooted in discrimination based on gender and citizenship.

Intersectionality emerges as a crucial determinant of wage outcomes. Non-citizen women experience a persistent additional penalty of 7.9% beyond what gender and citizenship status alone would predict, illustrating the concept of “double jeopardy” as described by Browne & Misra (2003). This compounded disadvantage aligns with Vernez’s (1999) findings on heightened cultural and institutional barriers uniquely faced by immigrant women.

Future research should explore credential devaluation by disaggregating foreign education by country or region, distinguishing, for example, the wage returns to Mexicans versus Southeast Asians degrees, the two largest immigrant groups in this study. Additionally, separating non-citizens by visa category (e.g. H1-B versus H2-A) could uncover heterogeneity in labor market outcomes obscured in aggregate analyses. Finally, targeted policy proposals—such as incentives for firms to adopt anti-bias training in high-immigrant sectors and the implementation of streamlined foreign credential recognition programs—deserve empirical evaluation, especially given the feasibility constraints imposed by current U.S. labor law.

The devaluation of female labor exposes the entrenched sexism and xenophobia that persist in the U.S. labor market. When these forces intersect, they compound wage penalties for immigrant-Latina workers. To address these disparities, policymakers must confront discrimination head-on. Structural reforms must go beyond improving access to human capital—they must dismantle the biases embedded in labor market institutions. Only then can we move beyond viewing immigrant labor as an *undesired necessity* and toward creating an equitable and inclusive labor market that benefits women immigrants and the U.S. economy, and not just the latter.

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VIII. Appendix

Table 1: Distribution of Citizenship Status in the Sample

citizenship status	Freq.	Percent	Cum.
citizen	448681	85.17	85.17
naturalized citizen	43450	8.25	93.41
not a citizen	34701	6.59	100.00
Total	526832	100.00	

Table 2: Annual Mean Wages by Sex and Citizenship

citizenship status	Sex		
	male	female	Total
citizen	63130.29	41446.54	52516.93
naturalized citizen	69754.73	47531.38	58556.11
not a citizen	48421.68	31863.99	41713.38
Total	62537.43	41440.69	52303.41

Table 3: Industry Distribution for Non-Citizens

Industry Name	Freq.	Percent	Cum.
Agriculture & Forestry	885	2.55	2.55
Arts	4601	13.26	15.81
Construction	3498	10.08	25.89
Education & Health	5669	16.34	42.23
Finance & Real Estate	1516	4.37	46.60
Information	477	1.37	47.97
Manufacturing	4867	14.03	62.00
Military	8	0.02	62.02
Mining & Oil	204	0.59	62.61
Other Services	1913	5.51	68.12
Professional Services	4779	13.77	81.89
Public Administration	395	1.14	83.03
Retail	3234	9.32	92.35
Transportation	1600	4.61	96.96
Wholesale	1055	3.04	100.00
Total	34701	100.00	

A. Detailed Blinder-Oaxaca Analysis¹¹

1. Gender Wage Decomposition

Table 5: Gender Oaxaca Decomposition

	Coefficient	Robust Std. Err	z	P>z	95% conf.	Interval
Male	10.576	0.002	5084.88	0	10.572	10.58
Female	10.179	0.002	4727.83	0	10.174	10.183
Wage Gap	0.397	0.003	132.63	0	0.391	0.403
Explained	-0.014	0.002	-7.57	0	-0.017	-0.01
Unexplained	0.392	0.002	166.93	0	0.387	0.397
Interaction	0.019	0.001	27.05	0	0.017	0.02

Table 5 presents the Blinder-Oaxaca decomposition results for the gender wage gap. The overall log wage difference between males and females is 0.397 (approximately 39.7%), consistent with the gender coefficient in the regression models. This substantial gap can be decomposed into its constituent parts:

The explained component is negative (-0.014), suggesting that women in the sample actually possess slightly better observable human capital characteristics than men—challenging the assumption that men are more qualified. However, this advantage is far outweighed by the unexplained component (0.392), which accounts for 98.7% of the total wage gap. This indicates that nearly the entire gender wage differential stems from differences in how the labor market rewards identical qualifications for men and women, rather than from differences in the qualifications themselves. Education contributes significantly to the unexplained portion (-0.254), indicating that women receive substantially lower returns to education than men with comparable credentials. Household income (0.106) and English proficiency (0.073) also exhibit

¹¹ All Oaxaca decompositions use robust standard errors, improving inference by correcting for heteroskedasticity. While coefficient estimates remain unchanged, some standard errors increased slightly, especially for interaction terms, making certain estimates (i.e. Non-Citizen Female vs. Citizen Female interaction) marginally less significant.

large unexplained effects, suggesting that these characteristics generate wage premiums for men that women do not receive equally. The constant term (0.457) captures the residual, unexplained baseline difference that persists even when all measured characteristics are equal—potentially reflecting pure discrimination or unobserved structural factors.

These findings support statistical discrimination theory, suggesting that employers systematically undervalue women’s productivity despite equivalent or superior qualifications.

2. Citizenship Status Decomposition

Table 6: Citizenship Oaxaca Decomposition

	Coefficient	Robust Std. Err	z	P>z	95% conf.	Interval
Citizens	10.399	0.002	6590.55	0	10.396	10.402
Not Citizens	10.156	0.006	1835.07	0	10.145	10.167
Wage Gap	0.243	0.006	42.21	0	0.232	0.254
Explained	0.207	0.009	24.370	0	0.191	0.224
Unexplained	0.011	0.005	1.980	0.048	0	0.021
Interaction	0.025	0.008	3.000	0.003	0.009	0.041

Table 6 presents the decomposition of the wage gap between citizens and non-citizens. The overall log wage difference is 0.243 (approximately 24.3%), with non-citizens earning less than citizens. Unlike the gender decomposition, the explained component (0.208) accounts for 85.6% of this gap, indicating that differences in observable characteristics—such as education and economic resources—explain the majority of the citizenship-based wage disparity.

The unexplained component (0.010) is small, positive, and not statistically significant, suggesting that citizens and non-citizens receive relatively similar returns to their characteristics. The interaction term (0.025) contributes modestly to the overall difference.

Education (0.093) and household income (0.099) are the leading contributors to the explained component, reflecting the greater human capital of citizens. Although the unexplained component for education (0.260) is positive—suggesting that citizens receive higher returns—it is partially offset by negative unexplained effects in other variables.

Overall, this decomposition suggests that the citizenship wage gap is driven primarily by structural inequalities in human capital rather than labor market discrimination, in contrast to the gender decomposition.

3. Non-Citizen Females vs. Citizen Females

Table 7: Non-Citizen Females vs Citizen Females Oaxaca Decomposition

	Coefficient	Robust Std. Err	z	P>z	95% conf.	Interval
Non-C Female	9.893	0.009	1117.46	0	9.875	9.91
Citizen Female	10.195	0.002	4603.83	0	10.191	10.2
Wage Gap	-0.302	0.009	-33.15	0	-0.32	-0.285
Explained	-0.195	0.007	-29.290	0	-0.207	-0.181
Unexplained	-0.086	0.015	-5.770	0	-0.117	-0.058
Interaction	-0.022	0.013	-1.630	0.104	-0.046	0.004

[Table 7](#) examines the wage gap between non-citizen and citizen females, offering insight into the specific effects of citizenship on women. The overall log wage difference is -0.302 (approximately 30.2%), with non-citizen women earning substantially less.

The explained component (-0.195) accounts for 64.6% of this gap, while the unexplained component (-0.086) explains 28.5%. The remaining 7.3% is captured by the interaction effect (-0.022). This suggests that while differences in observable characteristics are the primary driver, differential returns to those characteristics—likely stemming from discrimination—also play a significant role.

Education dominates both components, contributing -0.179 to the explained and -0.454 to the unexplained effect. This indicates that non-citizen women not only possess less education on average but also receive significantly lower returns on their educational attainment. The constant term (0.283) provides a partial counterbalance, but is not enough to offset the negative impacts.

These findings highlight the dual disadvantage faced by non-citizen women, who are penalized both for their lack of resources and for how those resources are valued by the labor market.

4. Non-Citizen Females vs. Non-Citizen Males

Table 8: Non-Citizen Females vs Non-Citizen Males Oaxaca Decomposition

	Coefficient	Robust Std. Err	z	P>z	95% conf.	Interval
Non-C Women	9.893	0.009	1117.46	0	9.875	9.91
Non-C Male	10.336	0.007	1517.75	0	10.322	10.349
Wage Gap	-0.443	0.011	-39.64	0	-0.465	-0.421
Explained	0.029	0.007	4.18	0	0.015	0.042
Unexplained	-0.476	0.009	-51.470	0	-0.494	-0.457
Interaction	0.004	0.002	1.69	0.091	-0.001	0.008

Table 8 presents perhaps the most revealing comparison, isolating gender differences within the non-citizen population. The overall log wage difference is -0.443 (approximately 44.3%), larger than the overall gender wage gap in the full sample.

Strikingly, the explained component (0.029) is positive, indicating that non-citizen women have slightly better observable characteristics than non-citizen men. However, the unexplained component (-0.476) is overwhelmingly negative, accounting for 107.4% of the total wage gap. The interaction effect (0.004) is minimal.

This stark unexplained gap is driven by large negative values associated with English proficiency (-0.089), household income (-0.124), and the constant term (-0.192). Although

education has a small positive unexplained effect (0.052), it is insufficient to offset the broader penalties.

These results indicate that non-citizen women face the most severe form of differential treatment in the labor market. Despite having slightly better characteristics than their male counterparts, they receive significantly lower returns, resulting in the largest wage gap of all groups examined.

5. Synthesis of Decomposition Findings

First, gender-based wage gaps are predominantly driven by unexplained differences, meaning the labor market assigns lower returns to women's qualifications than to men's. This pattern holds even within the non-citizen subgroup, underscoring systemic gender-based discrimination. Second, citizenship-based wage gaps are largely explained by observable human capital disparities, such as differences in education and household income. This suggests that citizenship status impacts wages more through structural disadvantages than through unequal treatment of similarly qualified individuals. Third, intersectionality amplifies these disadvantages. Non-citizen women face the "double penalty" observed in both regression and decomposition results: their wage gaps arise from both deficits in observable characteristics and from lower returns to those same characteristics. The gender penalty is especially pronounced within the non-citizen group, where women face a larger wage gap (44.3%) than in the full sample (39.7%).

Overall, these decomposition findings provide compelling evidence that the intersectional disadvantage experienced by non-citizen women stems from both structural barriers to human capital and discriminatory patterns in labor market valuation. These results align with statistical discrimination theory, where employers may use gender and citizenship as heuristics for

productivity—penalizing those who belong to marginalized groups even when qualifications are equal or superior.

B. Robustness Check:

Model 3 employs robust standard errors clustered at the geographic level (PUMA-state), ensuring valid inference across the 118 clusters. This approach accounts for spatial correlation in the data, which is particularly important given the large and geographically diverse sample. While I attempted additional robustness checks using wild cluster bootstrap procedures clustered by year, industry, and birthplace, these proved computationally infeasible due to the simultaneous inclusion of high-dimensional fixed effects for all three dimensions.

Despite these limitations, the core findings remain stable across specifications. While slight variations in coefficient estimates are observed across Models 1–3 in [Table 4](#) and [Table 9](#), the overall patterns remain consistent. In particular, both the gender wage gap and the intersectional penalty for non-citizen women are statistically significant at the 1% level. These small shifts reinforce the robustness of the study’s main results and support the validity of its conclusions.

Table 9: Effect of Gender and Citizenship on Log Wages

VARIABLES	Model 1	Model 2	Model 3
Naturalized Citizen (NC)	0.148*** (0.007)	0.011* (0.006)	0.052*** (0.017)
Not a Citizen (NotC)	-0.247*** (0.007)	-0.001 (0.007)	0.028 (0.018)
Female	-0.403*** (0.003)	-0.395*** (0.003)	-0.327*** (0.010)
NC Female	0.019* (0.010)	0.025 *** (0.008)	0.008 (0.018)
Not Citizen Female	-0.040*** (0.012)	-0.074*** (0.010)	-0.074*** (0.015)
Children: One		0.039*** (0.003)	0.041*** (0.004)
Children: Two		0.001 (0.003)	0.013** (0.006)
Children: Three-Four		-0.044*** (0.004)	-0.025*** (0.007)
Children: 5 or more		-0.104*** (0.015)	-0.081*** (0.016)
English: Intermediate		-0.156*** (0.007)	-0.132*** (0.009)
English: Limited		-0.305*** (0.007)	-0.271*** (0.013)
Education: Medium		0.205*** (0.003)	0.188*** (0.005)
Education: High		0.523*** (0.003)	0.473*** (0.011)
Race: Black		-0.091*** (0.004)	-0.124*** (0.009)
Race: American Indian/Alaska Native		-0.087*** (0.019)	-0.053* (0.029)
Race: Asian/Pacific Islander		-0.022*** (0.006)	-0.058*** (0.016)
Race: Hispanic/Latino		-0.003 (0.004)	-0.025*** (0.007)
Constant	10.583*** (0.002)	10.300*** (0.004)	10.276*** (0.013)
Observations	526,832	526,832	526,832
R-squared	0.038	0.423	0.457

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Note: Household income is included in Models 1 and 2, significant at the 1% level, but appears near zero due to scaling and is omitted in Model 3 due to collinearity with fixed effects. Age and age squared are included and significant in all models but excluded from the table for brevity. Controls and fixed effects match those in Table 4.